

Worksheet · Section 4.1

The Internet



The Big Picture

The Internet is a network of networks that runs on shared rules called protocols. Practice the vocabulary, how packets travel, Internet vs. Web, TCP vs. UDP, and bandwidth math.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|----------------------------------|---|
| _____ 1. Computer network | A. The maximum data sent per second |
| _____ 2. Internet | B. Faster protocol with no delivery guarantee |
| _____ 3. World Wide Web | C. The protocol the World Wide Web uses |
| _____ 4. Protocol | D. An agreed-upon set of rules for how systems communicate |
| _____ 5. Packet | E. A group of connected devices that can send and receive data |
| _____ 6. Router | F. A network of networks using open, standard protocols |
| _____ 7. Routing | G. Protocol that guarantees reliable, in-order delivery |
| _____ 8. Bandwidth | H. A system of linked pages and files that uses the Internet |
| _____ 9. IP address | I. A chunk of data plus metadata used to route it |
| _____ 10. TCP | J. A device that forwards packets toward their destination |
| _____ 11. UDP | K. Finding a path from sender to receiver |
| _____ 12. HTTP | L. A unique number that identifies a device on the Internet |

Part 2 · Fill It In

Use each word from the word bank once.

TCP**routing****packets****HTTP****UDP****IP****bandwidth****protocols**

1. Data is sent across the Internet broken into _____.
2. Agreed-upon rules for communication are _____.
3. Every device on the Internet is identified by an _____ address.
4. _____ reorders packets and resends lost ones.
5. _____ is faster but doesn't guarantee delivery.
6. The World Wide Web uses _____.
7. The maximum amount of data sent per second is _____.
8. Finding a path from sender to receiver is _____.

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Practice

Part 3 · Web or Just Internet?

Check whether each activity uses the World Wide Web or uses the Internet but not the Web.

Item	The Web	Internet (not Web)
1. Loading a website in a browser		
2. A multiplayer game sending players' moves		
3. A video call in a calling app		
4. Clicking a hyperlink		
5. A smart thermostat reporting the temperature		
6. Reading an online encyclopedia article		

Part 4 · TCP or UDP?

Activity	Protocol
Downloading a PDF	_____
Live video call	_____
Loading a web page	_____
Online racing game positions	_____
Sending an email	_____

Rule of thumb: if every bit must arrive, use _____; if speed matters more than a lost packet, use _____.

Part 5 · A Packet's Journey

Put the steps in order (write the letters):

- A. Routers forward packets, possibly along different paths
- B. The message is split into packets labeled with IP addresses and sequence numbers
- C. TCP puts packets back in order and requests missing ones
- D. The complete message is reassembled and shown

Order: _____ → _____ → _____ → _____

Why can packets arrive out of order?

Part 6 · Bandwidth Math

Remember: 1 byte = 8 bits. Show your work.

a) 100 megabits per second; 50 megabyte file → _____ seconds

b) 25 megabits per second; 1,000 megabyte file → _____ seconds

c) 200 megabits per second; 1,500 megabyte file → _____ seconds

Why multiply by 8? _____



Think:

Internet = the roads. Web = one kind of traffic on those roads. Email, games, and video calls use the Internet too, but they aren't the Web.

Name: _____

Class: _____

Date: _____

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is a protocol?
- A. A web page
 - B. A type of cable
 - C. A password
 - D. An agreed-upon set of rules for communication

★★ On Level

2. Why has the Internet's use of open protocols helped it grow?
- A. They prevent all errors
 - B. They make data private
 - C. Only one company can make devices
 - D. Any device or company can follow them to connect

★★ On Level

3. Which uses the Internet but is NOT part of the World Wide Web?
- A. Reading a web page
 - B. Viewing a website
 - C. A phone game sending scores to its server
 - D. Clicking a hyperlink

★★★ Exam Hard

4. A file is sent as 200 packets and 3 are lost. If the file is sent using TCP, what happens?
- A. The whole file is deleted
 - B. The file arrives with 3 holes
 - C. The packets are compressed
 - D. The missing packets are requested and resent

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)
- A. IPv6 provides more addresses than IPv4.
 - B. Packets from one message can take different routes.
 - C. Bandwidth is measured in bytes per web page.
 - D. The Internet is owned and run by one company.

Part 8 · Written Response

Explain, step by step, what happens when you load a web page. Use the words packet, IP address, router, TCP, and HTTP.
